Wormald's Tournament Conjectures 1, 2 and 6: Hessian Identities and Random Block Models

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21 October 2026. Public research manuscript, version 0.2.6.

Abstract

We study three specific questions in Wormald's random blow-up models. We express the logarithmic angular Hessian defined in that paper as a compression of a finite matrix, prove its positive definiteness, and identify the model's asymptotic constant with a determinant ratio. For a carousel base tournament, this ratio has a closed form. After multiplication by the factor for regular internal blocks, the resulting constant increases strictly to $\cosh(1)/\cos(1)$ and admits an explicit expansion in inverse powers of the base order. Combining the explicitly cited moment comparison of Fang and Huang with Wormald's model counting theorems, and treating nonregular bases separately, yields the unique maximizer of the expectations when the odd base order is fixed and the blocks are sufficiently large.

This manuscript addresses Conjectures 1, 2 and 6 in Wormald's public version. The derivations and verification programs were developed with AI assistance and await independent mathematical review.

Correspondence with the original conjectures and notation

We use N. C. Wormald's version arXiv:2610.16245v1 and restate three of its conjectures in the notation of this manuscript.

Fix an odd integer $d \geq 3$. The base tournament T^* has d vertices, each replaced by m vertices, giving total order $N = dm$. Directions between blocks are determined by T^*. In $\mathcal{T}(T^*, m)$ the internal blocks are independent uniform random tournaments. In $\mathcal{R}(T^*, m)$ each internal block is additionally regular, requiring m to be odd. Let $C(G)$ and $H(G)$ denote the numbers of Hamiltonian cycles and paths, respectively.

Definition 1 (Tournaments, counts and the carousel). A tournament orients every pair of distinct vertices in exactly one direction. Its adjacency entry in position i, j is 1 exactly when $i \to j$, and the diagonal is zero. In a regular tournament each vertex has outdegree and indegree $(d-1)/2$.

Hamiltonian paths are counted as directed vertex orderings. Hamiltonian cycles are identified up to cyclic starting point: rotating the starting point does not produce another cycle, and reversal is not an additional identification of directed orderings. An ordinary isomorphism is a vertex bijection preserving all edge directions. Every uniqueness assertion in this manuscript uses this notion.

For odd order, the carousel tournament Car_d is defined by

$$V(\mathrm{Car}_d) = \mathbb{Z}/d\mathbb{Z}, \qquad i \to j \iff (j-i) \bmod d \in \{1, \ldots, (d-1)/2\}.$$

- Conjecture 1: for fixed d and sufficiently large m, among bases of order d the carousel maximizes $\mathbb{E}C(\mathcal{T}(T^*, m))$. Part II proves this assertion and gives uniqueness, a path counterpart, and a counterpart with regular internal blocks.
- Conjecture 2: the limiting constant for a carousel base with regular internal blocks converges as odd $d \to \infty$. Part I identifies the limit and proves strict monotonicity and an error expansion.
- Conjecture 6: for each regular base of order at least three, the original Hessian determinant $\mathcal{H}$ is nonzero and $d/\mathcal{H}$ is a rational square. Part I gives exact formulas.

The original Conjecture 3 concerns the exact cycle extremum among all tournaments of sufficiently large odd order. It lies outside this manuscript. We also do not claim to prove the global statements of the original Conjectures 4 and 5.

Reading route and external inputs

Sections 1-7 of Part I use finite dimensional linear algebra and differentiation. Section 8 invokes Wormald's Theorems 1, 4 and 5 to connect the finite Hessian with asymptotic expectations in the random models. Sections 9-11 compute the carousel closed form and its expansion.

Section 3 of Part II uses Fang-Huang's Corollary 3.4. The equality classification for the fourth moment can also be checked directly through the four-vertex calculation in Appendix A.3. Section 8 of Part II uses Brégman's permanent bound for zero-one matrices. Nonregular bases are treated separately, so the regular Hessian formula is never applied outside its hypotheses.

The appendices give the determinant polynomial, carousel isomorphism, fourth-moment equality, termination of row integrality, path normalization and common finite threshold. A separate elementary guide is provided in Chinese.

Every limit $m \to \infty$ in this manuscript fixes d first. The subsequent discussion of model constants as $d \to \infty$ does not interchange limits and does not give a uniform second-order error in the final order N.

Part I: The exact Hessian and the model constants

1. Objects, main formulas and scope

Write I_r for the identity matrix of order r, J_r for the all-one matrix of order r, $\mathbf{1}_r$ for the all-one column vector. Matrix subscripts may be omitted when multiplication determines their dimensions. Superscript T denotes transpose, and the imaginary unit satisfies $i^2 = -1$.

Definition 2 (Sign matrix and original angular function). Let $d \geq 3$ be odd and let A be the adjacency matrix of a regular tournament of order d.

$$S = 2A + I - J, \quad S^T = -S, \quad S\mathbf{1} = 0, \qquad C = S/d, \quad P_0 = J/d.$$

The matrix $\operatorname{diag}(a_1, \ldots, a_r)$ has the listed scalars on the diagonal and zero elsewhere.

Set

$$B(y) = I - (A + \tfrac{1}{2}I)\operatorname{diag}(y_1, \ldots, y_d), \quad K(y) = \det B(y), \quad z = 2/d.$$

On positive real numbers, log is the natural logarithm; in complex neighborhoods we use the analytic branches specified below.

Near $y = z\mathbf{1}$, Wormald solves $K(y) = 0$ for $y_d = h(y_1, \ldots, y_{d-1})$ and takes the Hessian of

$$\widetilde{f}(\theta) = \log h(ze^{i\theta_1}, \ldots, ze^{i\theta_{d-1}})$$

Write e_j for the jth standard basis column vector.

On a graph, $C(G), H(G)$ count cycles and paths. In contrast, the standalone $C = S/d$ is a matrix, bold $\mathbf{H}$ is the Hessian matrix, and $\mathcal{H}$ is its determinant. Local symbols in different sections are interpreted according to the definitions given in those sections.

Theorem 3 (The original Hessian and the positive rational square). Evaluate this Hessian at $\theta = 0$. Denote its matrix by $\mathbf{H}$ and its determinant by $\mathcal{H}$. Sections 4-6 verify the conditions needed to define h and the logarithm branch, so this definition does not assume an existence statement still to be proved. We prove

$$\boxed{\mathbf{H} = E^T(I + C^2)(I - C^2)^{-1}E, \qquad E = (e_1 - e_d, \ldots, e_{d-1} - e_d).}$$

Consequently,

$$\boxed{\mathcal{H} = d\,\frac{\det(I + C^2)}{\det(I - C^2)} > 0, \qquad \sqrt{\frac{d}{\mathcal{H}}} = \frac{\det(I + C)}{\det(I + iC)} =: \rho_d(S).}$$

The right-hand side is positive and rational, so these identities prove Conjecture 6.

2. Spectral gap, the constant direction and restricted determinants

Proof. (Hessian identity)

Throughout, $\mathbf{1}$ denotes the all-one column vector. The tournament definition and regularity give

$$A + A^T = J - I, \qquad A\mathbf{1} = \frac{d-1}{2}\mathbf{1},$$

$$S^T = 2A^T + I - J = 2(J - I - A) + I - J = -S,$$

$$S\mathbf{1} = 2\frac{d-1}{2}\mathbf{1} + \mathbf{1} - d\mathbf{1} = 0.$$

The matrix S has $d(d-1)$ off-diagonal entries, all equal to ± 1. Thus

For a real vector $v = (v_1, \ldots, v_r)^T$, its Euclidean norm is $\|v\|_2 = (\sum_{j=1}^r v_j^2)^{1/2}$.

For a real matrix M, the two norms are

$$\|M\|_F^2 = \sum_{j,k} M_{jk}^2, \qquad \|M\|_{\mathrm{op}} = \sup_{v \neq 0} \frac{\|Mv\|_2}{\|v\|_2}.$$

Lowercase letters may denote entries, for example $s_{ij} = S_{ij}$.

$$\|S\|_F^2 = \sum_{i \neq j} s_{ij}^2 = d(d-1), \qquad \|C\|_F^2 = \frac{d-1}{d}.$$

Apply a real orthogonal basis change to the normalized skew-symmetric matrix C. Its canonical form consists of zero blocks and two-dimensional blocks

$$J_\lambda = \begin{pmatrix} 0 & \lambda \\ -\lambda & 0 \end{pmatrix}$$

Each two-dimensional block contributes $2\lambda^2$ to the squared Frobenius norm. Therefore every frequency satisfies

$$2\lambda^2 \le \sum_{\text{all two-dimensional blocks}} 2\lambda_j^2 = \|C\|_F^2 = \frac{d-1}{d},$$

$$\|C\|_{\text{op}}^2 = \max_j \lambda_j^2 \le \frac{d-1}{2d} < \frac{1}{2}.$$

Put $u = \mathbf{1}/\sqrt{d}$, $P_0 = uu^T = J/d$ and $\Pi = I - P_0$. Direct multiplication gives

$$u^T u = 1, \quad P_0^2 = P_0, \quad \Pi^2 = I - 2P_0 + P_0^2 = \Pi,$$

$$Cu = 0, \qquad u^T C = -(Cu)^T = 0, \qquad CP_0 = (Cu)u^T = 0, \quad P_0 C = u(u^T C) = 0.$$

We also have

$$P := \frac{2A + I}{d} = P_0 + C, \qquad Pu = P_0 u + Cu = u.$$

If $u^T v = 0$, then $P_0 v = u(u^T v) = 0$, and hence $Pv = Cv$. At the critical point,

$$B(z\mathbf{1}) = I - P = \Pi - C, \qquad B(z\mathbf{1})u = 0.$$

The space $u^\perp$ is orthogonal to the real vector u. The notation $\det_{u^\perp}$ means the determinant of the restriction to this invariant subspace, computed in an orthonormal basis.

On $u^\perp$ this matrix is $I - C$. If $x \in u^\perp$ satisfies $(I - C)x = 0$, then

$$0 = x^T(I - C)x = \|x\|_2^2 - x^T C x = \|x\|_2^2,$$

where $x^T C x = -x^T C x$ implies $x^T C x = 0$. Thus $x = 0$, and the restricted matrix is invertible.

On a two-dimensional block,

$$\det(I - J_\lambda) = \det \begin{pmatrix} 1 & -\lambda \\ \lambda & 1 \end{pmatrix} = 1 + \lambda^2 > 0.$$

The eigenvalue of $I - C$ in the constant direction is 1. Therefore

$$q := \det_{u^\perp}(I - C) = \det(I - C) = \prod_j (1 + \lambda_j^2) > 0.$$

Taking transposes gives

$$\det(I + C) = \det((I + C)^T) = \det(I - C) = q.$$

3. Logarithmic variables, block matrices and Schur elimination

In this part $1 \le j, k \le d$ until Section 6 eliminates the last coordinate and restricts them to $1 \le j, k \le d - 1$. Take $t = (t_1, \dots, t_d) \in \mathbb{C}^d$ near the origin and write $y = (y_1, \dots, y_d)$. The angular variables are $\theta = (\theta_1, \dots, \theta_{d-1}) \in \mathbb{R}^{d-1}$. Set $y_j = ze^{t_j}$. Then

$$\mathcal{B}(t) = B(ze^{t_1}, \dots, ze^{t_d}) = I - P\,\mathrm{diag}(e^{t_1}, \dots, e^{t_d}).$$

Choose an orthonormal basis of $u^{\perp}$ as the columns of V. The matrix $O = (u, V)$ satisfies

$$V^T V = I_{d-1}, \quad V^T u = 0, \quad VV^T = \Pi, \quad O^T O = OO^T = I_d.$$

In this basis write

$$O^T \mathcal{B}(t) O = \begin{pmatrix} a(t) & b(t) \\ c(t) & D(t) \end{pmatrix},$$

$$a = u^T \mathcal{B} u, \quad b = u^T \mathcal{B} V, \quad c = V^T \mathcal{B} u, \quad D = V^T \mathcal{B} V.$$

Here a is a scalar, b is a row of size $1 \times (d-1)$, c is a column of size $(d-1) \times 1$, and D is square of order $(d-1)$. The derivatives below also use the left constant-vector relation, which follows from skew symmetry as

$$u^T P = u^T (P_0 + C) = u^T u u^T + u^T C = u^T + 0 = u^T.$$

Since $\det(O)^2 = 1$, this basis change preserves the determinant. At $t = 0$,

$$a(0) = 0, \quad b(0) = 0, \quad c(0) = 0, \qquad D(0) = V^T (I - C) V.$$

Section 2 proves that $D(0)$ is invertible and that $\det D(0) = q > 0$. Continuity of the determinant gives a neighborhood of the origin in which $\det D(t) \neq 0$. The inverse formula

For a square matrix D, the adjugate $\operatorname{adj} D$ is the transpose of the cofactor matrix: $(\operatorname{adj} D)_{ij} = (-1)^{i+j} \det D^{(j|i)}$, where $D^{(j|i)}$ deletes row j and column i. Determinant expansion gives $D \operatorname{adj} D = (\det D) I$. When $\det D \neq 0$, division by this determinant gives the inverse formula used below.

$$D(t)^{-1} = \frac{\operatorname{adj} D(t)}{\det D(t)}$$

shows that every entry of the inverse is analytic on this neighborhood.

The actual block elimination is

$$\begin{pmatrix} 1 & -bD^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} a & b \\ c & D \end{pmatrix} = \begin{pmatrix} a - bD^{-1}c & 0 \\ c & D \end{pmatrix}.$$

The left multiplier has determinant 1, and the resulting matrix is block triangular. Defining

$$\psi(t) = a(t) - b(t) D(t)^{-1} c(t)$$

therefore gives the exact identity

$$\det \mathcal{B}(t) = \det D(t)\, \psi(t).$$

On this neighborhood, $\det \mathcal{B}(t) = 0$ and $\psi(t) = 0$ are equivalent because the factor $\det D(t)$ is nonzero.

4. First and second derivatives: all nine product terms

For a scalar function $F(t_1,\dots,t_d)$, subscripts F_j, F_{jk} denote $\partial F/\partial t_j$ and $\partial^2 F/(\partial t_j \partial t_k)$; matrix, row and column derivatives are taken entry by entry. When differentiating in the original variables y, we write K_{y_j}. The matrix $\nabla^2 F$ consists of the entries F_{jk}.

The symbol δ_{jk} is 1 when $j = k$ and 0 otherwise.

Write $F_j = e_j e_j^T$ and $R(t) = D(t)^{-1}$. Only column j of the exponential matrix depends on t_j, so at the origin

$$\mathcal{B}_j = -PF_j, \qquad \mathcal{B}_{jk} = -\delta_{jk} PF_j.$$

To make the inverse derivatives explicit, differentiate $DR = I$ once and then again:

$$D_j R + DR_j = 0, \qquad R_j = -RD_j R,$$

$$R_{jk} = RD_k RD_j R - RD_{jk} R + RD_j RD_k R.$$

These identities hold in the neighborhood where $\det D \neq 0$. The symbols R_j, R_{jk} in the nine terms below denote these derivatives.

Using $u^T P = u^T$ and $e_j^T u = 1/\sqrt{d}$,

$$a_j(0) = -u^T PF_j u = -u_j^2 = -1/d, \qquad a_{jk}(0) = -\delta_{jk}/d.$$

First list every term of the first product derivative:

$$\partial_j (bRc) = b_j Rc + bR_j c + bRc_j.$$

Differentiating with respect to t_k gives all nine terms:

$$\begin{aligned}\partial_k \partial_j (bRc) = {} & b_{jk} Rc + b_j R_k c + b_j Rc_k \\ & + b_k R_j c + bR_{jk} c + bR_j c_k \\ & + b_k Rc_j + bR_k c_j + bRc_{jk}.\end{aligned}$$

At the origin $b = c = 0$. Terms 1, 2, 4, 5, 6, 8 and 9 contain a factor b or c and hence vanish there. The two surviving terms are

$$\partial_k \partial_j (bRc)(0) = b_j(0) R(0) c_k(0) + b_k(0) R(0) c_j(0).$$

We now calculate these factors:

$$b_j(0) = -u^T Pe_j e_j^T V = -\frac{1}{\sqrt{d}} e_j^T V,$$

$$c_k(0) = -V^T Pe_k e_k^T u = -\frac{1}{\sqrt{d}} V^T Ce_k,$$

where $V^T P = V^T C$ follows from $V^T u = 0$. Set $U = (I - C)^{-1}$. The matrix C preserves $u^\perp$, and $D(0) = V^T (I - C) V$ gives the identities below. Multiply the first on the left by U and on the right by $R(0)$ to obtain the second; the next line uses $\Pi C = C$:

$$(I-C)V = VD(0), \qquad UV = VR(0),$$

$$UC = U\Pi C = UVV^TC = VR(0)V^TC.$$

Consequently,

$$VR(0)V^TC = UC.$$

The two second-order products are therefore

$$b_jR(0)c_k = \frac{1}{d}e_j^TUCe_k = \frac{(UC)_{jk}}{d},$$

$$b_kR(0)c_j = \frac{1}{d}e_k^TUCe_j = \frac{(UC)_{kj}}{d}.$$

The first derivative of the product vanishes at the origin. Thus

$$\psi_j(0) = -1/d, \qquad \psi_{jk}(0) = -\frac{\delta_{jk} + (UC)_{jk} + (UC)_{kj}}{d}.$$

From $C(I-C) = (I-C)C$, multiplying on the left and right by the inverse gives $CU = UC$. Also,

$$(UC)^T = C^TU^T = -C(I+C)^{-1}.$$

The difference of the two inverses expands as

$$\begin{aligned}(I-C)^{-1} - (I+C)^{-1} &= (I-C)^{-1}[(I+C) - (I-C)](I+C)^{-1} \\ &= 2C(I-C)^{-1}(I+C)^{-1} \\ &= 2C(I-C^2)^{-1}.\end{aligned}$$

It follows that

$$\begin{aligned}I + UC + (UC)^T &= I + C[(I-C)^{-1} - (I+C)^{-1}] \\ &= I + 2C^2(I-C^2)^{-1} \\ &= (I+C^2)(I-C^2)^{-1} =: Q.\end{aligned}$$

We have obtained the full derivative formulas

$$\boxed{\psi_j(0) = -1/d, \qquad \nabla^2\psi(0) = -Q/d.}$$

Differentiate the determinant product. Since $\psi(0) = 0$ and $\det D(0) = q$,

$$\begin{aligned}\partial_{t_j} \det \mathcal{B}(0) &= \partial_{t_j}(\det D)(0)\,\psi(0) + \det D(0)\psi_j(0) \\ &= 0 + q(-1/d) = -q/d.\end{aligned}$$

At the origin $\partial y_j/\partial t_j = z$, so

$$zK_{y_j}(z\mathbf{1}) = -q/d, \qquad K_{y_j}(z\mathbf{1}) = -\frac{q}{dz} = -q/2 \neq 0.$$

This supplies the nonzero partial derivative needed for every choice of eliminated coordinate.

5. Eliminating the last variable by an explicit rational function

The notations $y' = (y_1, \ldots, y_{d-1})$ and $t' = (t_1, \ldots, t_{d-1})$ delete the last coordinate; those primes do not denote derivatives.

The determinant is linear in column d, and this column is affine in y_d. Hence

$$K(y) = K_0(y_1, \ldots, y_{d-1}) + y_d K_1(y_1, \ldots, y_{d-1}).$$

The two coefficients have the explicit definitions

$$K_0(y') = K(y', 0), \qquad K_1(y') = K(y', 1) - K(y', 0) = K_{y_d}(y', y_d).$$

The last partial derivative does not depend on y_d, because the displayed expression is affine in y_d.

The derivative calculation in Section 4 gives

$$K_1(z, \ldots, z) = K_{y_d}(z\mathbf{1}) = -q/2 \neq 0.$$

Shrink the neighborhood so that $K_1(y') \neq 0$. We can then write explicitly

$$h(y') = -\frac{K_0(y')}{K_1(y')}.$$

Since $K(z\mathbf{1}) = 0$,

$$K_0(z, \ldots, z) + zK_1(z, \ldots, z) = 0, \qquad h(z, \ldots, z) = z.$$

Define

$$g(t') = \log\left[\frac{h(ze^{t_1}, \ldots, ze^{t_{d-1}})}{z}\right].$$

At the origin the logarithm's argument is 1. Continuity of h, together with $h(z, \ldots, z) = z > 0$, lets us shrink the neighborhood so that $|h(ze^{t'})/z - 1| < 1/2$, where $ze^{t'}$ means coordinatewise exponentiation. This disk excludes 0. Choose its analytic logarithm branch with value 0 at 1. Then

$$g(0) = \log 1 = 0, \qquad t_d = g(t'), \qquad \psi(t', g(t')) = 0.$$

We differentiate only this composition of an explicit rational function and an analytic logarithm; existence of an implicit function is not an additional assumption.

6. The complete chain rule and the angular sign

Here $j, k \in \{1, \dots, d-1\}$. The chain rule holds first in the surface neighborhood. The derivative values calculated in Section 4 are substituted only at $t' = 0,\ t_d = g(0) = 0$. In particular, $g_j(0) = -1$ is not an identity throughout that neighborhood. For $j \le d-1$, take one derivative:

$$0 = \psi_j + \psi_d g_j, \qquad g_j(0) = -\frac{\psi_j(0)}{\psi_d(0)} = -\frac{-1/d}{-1/d} = -1.$$

Differentiating again with respect to t_k gives five terms:

$$0 = \psi_{jk} + \psi_{jd} g_k + \psi_{dk} g_j + \psi_{dd} g_j g_k + \psi_d g_{jk}.$$

Substitute the derivatives from Section 4 and $g_j = g_k = -1$ term by term:

$$0 = -\frac{Q_{jk}}{d} + \frac{Q_{jd}}{d} + \frac{Q_{dk}}{d} - \frac{Q_{dd}}{d} - \frac{g_{jk}}{d}.$$

Multiply by d and isolate the last derivative:

$$\begin{aligned} g_{jk}(0) &= -Q_{jk} + Q_{jd} + Q_{dk} - Q_{dd} \\ &= -(e_j - e_d)^T Q (e_k - e_d). \end{aligned}$$

Choose the same local branch for the original $\log h$; when $h = z$, take the real value $\log z$. By the definition of g in Section 5, the original angular function satisfies

$$\widetilde{f}(\theta) = \log z + g(i\theta_1, \dots, i\theta_{d-1}).$$

Its first angular derivative is $i g_j(i\theta)$ and its second angular derivative is $i^2 g_{jk}(i\theta)$. Therefore

$$\mathbf{H}_{jk} = i^2 g_{jk}(0) = -g_{jk}(0) = (e_j - e_d)^T Q (e_k - e_d),$$

$$\boxed{\mathbf{H} = E^T Q E.}$$

7. Positive definiteness, the basis-change factor and the rational square

The notation $M \succ 0$ means that M is real symmetric and, for every nonzero real vector v, we have $v^T M v > 0$.

On each two-dimensional block from Section 2, $J_\lambda^2 = -\lambda^2 I_2$. Therefore

$$(I + J_\lambda^2)(I - J_\lambda^2)^{-1} = \frac{1 - \lambda^2}{1 + \lambda^2} I_2.$$

The coefficient is positive because $\lambda^2 < 1/2$. On zero blocks, Q is the identity, so $Q \succ 0$. For $v \in \mathbb{R}^{d-1}$,

$$Ev = (v_1, \dots, v_{d-1}, -\textstyle\sum_{j=1}^{d-1} v_j)^T.$$

If $v \neq 0$, its first $d-1$ components are not all zero, so $Ev \neq 0$. Consequently,

$$v^T \mathbf{H} v = (Ev)^T Q (Ev) > 0.$$

We next retain the volume factor of the basis change. For E, the inner product of columns j, k is

$$(e_j - e_d)^T (e_k - e_d) = \delta_{jk} + 1.$$

Thus $E^T E = I_{d-1} + J_{d-1}$. Its eigenvalue in the all-one direction is $1 + (d-1) = d$; each direction orthogonal to the all-one vector has eigenvalue 1. Therefore

$$\det(E^T E) = d \cdot 1^{d-2} = d.$$

Write $E = VL$, where V is an orthonormal basis of $u^\perp$ and L is invertible:

$$E^T Q E = L^T (V^T Q V) L, \qquad E^T E = L^T L.$$

Taking determinants gives, respectively,

$$\det(E^T Q E) = \det(L)^2 \det(V^T Q V), \qquad \det(L)^2 = d.$$

Using $Cu = 0$,

$$Qu = (I + C^2)(I - C^2)^{-1} u = u.$$

In the real orthogonal basis of Section 2, every block of Q is a real scalar multiple of the identity. Thus Q is real symmetric, and $u^T Q V = (Qu)^T V = u^T V = 0$. Hence

$$(u, V)^T Q (u, V) = \begin{pmatrix} 1 & 0 \\ 0 & V^T Q V \end{pmatrix}, \qquad \det Q = \det(V^T Q V).$$

Restore the two determinant factors:

$$\mathcal{H} = d \det Q = d \, \frac{\det(I + C^2)}{\det(I - C^2)} > 0.$$

Matrix multiplication and transposition give

$$(I + iC)(I - iC) = I + C^2, \qquad \det(I - iC) = \det((I - iC)^T) = \det(I + iC),$$

$$(I + C)(I - C) = I - C^2, \qquad \det(I - C) = \det(I + C).$$

Therefore

$$\det(I + C^2) = \det(I + iC)^2, \qquad \det(I - C^2) = \det(I + C)^2.$$

The sign of the square root is also fixed block by block:

$$\det(I + J_\lambda) = 1 + \lambda^2 > 0, \qquad \det(I + iJ_\lambda) = 1 - \lambda^2 > 0.$$

Thus

$$\sqrt{\frac{d}{\mathcal{H}}} = \frac{\det(I + C)}{\det(I + iC)} > 0.$$

Write $[d] = \{1, \dots, d\}$. The principal submatrix $M[U]$ uses the rows and columns in U, with convention $\det M[\varnothing] = 1$.

Finally, prove rationality. Expand the determinant multilinearly, choosing either an identity column or a column of C:

Write $\lfloor a \rfloor$ for the largest integer at most a.

$$\det(I + tC) = \sum_{U \subseteq [d]} t^{|U|} \det C[U].$$

For odd $r = |U|$,

$$\det C[U] = \det(C[U]^T) = \det(-C[U]) = (-1)^r \det C[U] = -\det C[U],$$

so this principal determinant is zero. Since C has rational entries,

$$\det(I + iC) = \sum_{k=0}^{\lfloor d/2 \rfloor} \sum_{\substack{U \subseteq [d] \\ |U| = 2k}} (-1)^k \det C[U] \in \mathbb{Q}.$$

The number $\det(I + C)$ is rational as well. We have already shown that numerator and denominator are positive, so their ratio is a positive rational number. □

8. External counting theorems and the four limits

Set $0! = 1$ and $n! = 1 \cdot 2 \cdots n$.

A vertex of type j belongs to the block replacing base vertex j, which contains m vertices.

Definition 4 (Normalization of the random models). This section explicitly invokes Wormald's Theorems 1, 4 and 5; it does not claim to reprove their random counting arguments. Fix a regular base of order d, internal block order m and total order $N = dm$. Subscripts $\mathcal{T}, \mathcal{R}$ specify the random model; for example, $\mathbb{E}C_{\mathcal{T}}$ is the expected cycle count of a random graph drawn from $\mathcal{T}(T^*, m)$. The path notation has the corresponding meaning. Blocks are independent and uniform over all admissible labeled internal tournaments. Define

$$\nu_N = \frac{(N-1)!}{2^N}, \qquad \mu_N = \frac{N!}{2^{N-1}}.$$

Corollary 5 (The four expectation limits). When $\mathcal{H} \neq 0$, Wormald's Theorem 4 gives

$$\frac{\mathbb{E}C_{\mathcal{T}}}{\nu_N} \longrightarrow \sqrt{d/\mathcal{H}}.$$

Section 7 establishes $\mathcal{H} > 0$ and the positive branch of its square root. Substitution gives

$$\lim_{m \to \infty} \frac{\mathbb{E}C_{\mathcal{T}}}{\nu_N} = \sqrt{\frac{d}{d \det(I + C^2)/\det(I - C^2)}} = \rho_d(S).$$

Theorem 5 replaces cycle counts in Theorem 4 by path counts. Thus

$$\lim_{m\to\infty} \frac{\mathbb{E}H_{\mathcal{T}}}{\mu_N} = \sqrt{d/\mathcal{H}} = \rho_d(S).$$

Theorem 1 gives the expectation ratio between regular internal blocks and independent random internal blocks:

$$\frac{\mathbb{E}C_{\mathcal{R}}}{\mathbb{E}C_{\mathcal{T}}} \longrightarrow e^{1/d}.$$

For sufficiently large m the denominator is positive, because its normalized limit is $\rho_d(S) > 0$. Multiplication gives

$$\lim_{\substack{m\to\infty \\ m \text{ odd}}} \frac{\mathbb{E}C_{\mathcal{R}}}{\nu_N} = \lim_{\substack{m\to\infty \\ m \text{ odd}}} \left(\frac{\mathbb{E}C_{\mathcal{R}}}{\mathbb{E}C_{\mathcal{T}}} \frac{\mathbb{E}C_{\mathcal{T}}}{\nu_N} \right) = e^{1/d}\rho_d(S).$$

Theorem 5 also replaces the counts in Theorem 1 by path counts, giving

$$\frac{\mathbb{E}H_{\mathcal{R}}}{\mathbb{E}H_{\mathcal{T}}} \longrightarrow e^{1/d}.$$

The fourth limit is therefore

$$\lim_{\substack{m\to\infty \\ m \text{ odd}}} \frac{\mathbb{E}H_{\mathcal{R}}}{\mu_N} = \lim_{\substack{m\to\infty \\ m \text{ odd}}} \left(\frac{\mathbb{E}H_{\mathcal{R}}}{\mathbb{E}H_{\mathcal{T}}} \frac{\mathbb{E}H_{\mathcal{T}}}{\mu_N} \right) = e^{1/d}\rho_d(S).$$

All four limits fix d. The last two take only admissible odd m and do not involve simultaneous growth of d and m.

9. A complete closed form for the carousel base

Theorem 6 (Closed forms for the carousel model). Let T_d have entries 1 above the diagonal, -1 below it and 0 on it. Appendix A.2 gives the signed permutation similarity between a carousel sign matrix and T_d. Appendix A.1 proves the determinant closed form and then extracts the appropriate coefficient to show that every even principal determinant is 1 and every odd principal determinant is 0. Hence

$$\det(I + tT_d) = \frac{(1+t)^d + (1-t)^d}{2}.$$

When a graph name is used as the argument of ρ_d below, it denotes that graph's sign matrix. Ordinary relabeling preserves the determinants, so the value does not depend on vertex labels. Consequently,

$$\boxed{\rho_d(\mathrm{Car}_d) = \frac{(1+1/d)^d + (1-1/d)^d}{(1+i/d)^d + (1-i/d)^d}.}$$

This directly recovers $2, 44/19, 3032/1231, \ldots$ in the original table, without calculating a multivariate Hessian separately for each example. The constant for regular internal blocks is

$$\mathfrak{r}_d = e^{1/d} r_d, \qquad r_d := \rho_d(\mathrm{Car}_d).$$

10. Real-function form and strict monotonicity

A prime on a function of one variable denotes differentiation in its argument.

Definition 7 (Real functions and auxiliary variables). The functions used here and in the Taylor calculation are defined by the following formulas; sec is used only where the cosine is nonzero.

$$\cosh y = \frac{e^y + e^{-y}}{2}, \quad \sinh y = \frac{e^y - e^{-y}}{2}, \quad \tanh y = \frac{\sinh y}{\cosh y}, \quad \operatorname{sech} y = \frac{1}{\cosh y}, \quad \sec y = \frac{1}{\cos y}.$$

Set $x = 1/d$ and define on $0 < x \le 1/3$

$$\alpha(x) = \frac{\arctan x}{x}, \qquad \beta(x) = \frac{\operatorname{artanh} x}{x}, \qquad \operatorname{artanh} x = \frac{1}{2}\log\frac{1+x}{1-x}.$$

Proof. (Strict monotonicity)

First express the numerator and denominator as real functions. The sum and difference of the real logarithms give

$$\log(1 \pm x) = \frac{1}{2}\log(1 - x^2) \ \pm \operatorname{artanh} x.$$

Exponentiating and adding gives

$$(1+x)^{1/x} + (1-x)^{1/x} = 2(1-x^2)^{1/(2x)} \cosh \beta(x).$$

For $0 \le x \le 1/3$, both $1 \pm ix$ have real part 1, so their entire curves lie in the right half-plane. Choose on that half-plane the analytic logarithm agreeing with the real logarithm on the positive axis. Then

$$\log(1 \pm ix) = \frac{1}{2}\log(1 + x^2) \pm i \arctan x,$$

$$(1+ix)^{1/x} + (1-ix)^{1/x} = 2(1+x^2)^{1/(2x)} \cos \alpha(x).$$

Therefore

$$\mathfrak{r}(x) = e^{p(x)}\frac{\cosh \beta(x)}{\cos \alpha(x)}, \qquad p(x) = x + \frac{\log(1-x^2) - \log(1+x^2)}{2x}.$$

When $x = 1/d$, the exponent $1/x = d$ is an integer, so these complex powers equal the original polynomial powers. Also $0 < \alpha(x) \le 1 < \pi/2$, so the cosine in the denominator is strictly positive.

Subtract the two logarithmic series:

$$\log(1-x^2) - \log(1+x^2) = -2\sum_{j=0}^{\infty} \frac{x^{4j+2}}{2j+1},$$

$$p(x) = x - \sum_{j=0}^{\infty} \frac{x^{4j+1}}{2j+1} = -\sum_{j=1}^{\infty} \frac{x^{4j+1}}{2j+1},$$

$$p'(x) = -\sum_{j=1}^{\infty} \frac{4j+1}{2j+1} x^{4j} < 0.$$

Changing variables in the integrals for the arctangent and inverse hyperbolic tangent gives

$$\alpha(x) = \int_0^1 \frac{dt}{1 + x^2 t^2}, \qquad \beta(x) = \int_0^1 \frac{dt}{1 - x^2 t^2}.$$

Set $v = 1 - \alpha$ and differentiate under the integral:

$$\beta'(x) = \int_0^1 \frac{2xt^2}{(1 - x^2 t^2)^2}\, dt, \qquad v'(x) = \int_0^1 \frac{2xt^2}{(1 + x^2 t^2)^2}\, dt > 0.$$

For each $t \in [0, 1]$,

$$\frac{2xt^2}{(1 - x^2 t^2)^2} \le \left(\frac{1 + x^2}{1 - x^2}\right)^2 \frac{2xt^2}{(1 + x^2 t^2)^2},$$

$$\left(\frac{1 + x^2}{1 - x^2}\right)^2 \le \left(\frac{1 + 1/9}{1 - 1/9}\right)^2 = \left(\frac{5}{4}\right)^2 = \frac{25}{16}.$$

Hence $\beta'(x) \le (25/16) v'(x)$. Since $\beta'(x) > 0$,

$$\beta(x) \le \beta(1/3) = 3 \cdot \frac{1}{2} \log \frac{1 + 1/3}{1 - 1/3} = \frac{3}{2} \log 2,$$

$$\tanh \beta(x) \le \frac{e^{3 \log 2} - 1}{e^{3 \log 2} + 1} = \frac{8 - 1}{8 + 1} = \frac{7}{9}.$$

Use $(1 + z)^{-1} \ge 1 - z$ and integrate:

$$\alpha(x) \ge \int_0^1 (1 - x^2 t^2)\, dt = 1 - \frac{x^2}{3} \ge \frac{26}{27}.$$

Appendix A.7 proves $\tan y \ge y + y^3/3$ by elementary calculus. Since $y + y^3/3$ is strictly increasing on nonnegative real numbers, substitute the lower bound $\alpha(x) \ge 26/27$ to obtain

$$\tan \alpha(x) \ge \frac{26}{27} + \frac{1}{3} \left(\frac{26}{27}\right)^3.$$

Putting the terms over a common denominator gives the following strict margin:

$$\frac{26}{27} + \frac{26^3}{3 \cdot 27^3} - \frac{7}{9}\frac{25}{16} = \frac{1191008 - 1148175}{944784} = \frac{42833}{944784} > 0.$$

Finally differentiate $\log \mathfrak{r}$ and substitute each of the bounds above:

$$\begin{aligned} \frac{d}{dx} \log \mathfrak{r}(x) &= p'(x) + \tanh \beta(x)\, \beta'(x) - \tan \alpha(x)\, v'(x) \\ &\le p'(x) + \left(\frac{7}{9}\frac{25}{16} - \frac{26}{27} - \frac{26^3}{3 \cdot 27^3}\right) v'(x) \\ &= p'(x) - \frac{42833}{944784} v'(x) < 0. \end{aligned}$$

Theorem 8 (Monotonicity and the limit). The inequality $d_1 < d_2$ implies $1/d_1 > 1/d_2$, hence $\mathfrak{r}_{d_1} < \mathfrak{r}_{d_2}$. Define $L := \cosh 1/\cos 1$. As $x \downarrow 0$,

$$\alpha(x) \to 1, \qquad \beta(x) \to 1, \qquad p(x) \to 0,$$

$$\lim_{d\to\infty,\ d \text{ odd}} \mathfrak{r}_d = e^0 \frac{\cosh 1}{\cos 1} = L.$$

11. Recovering every Taylor coefficient and remainder

In the expansions here, $R(x) = O(x^k)$ means that constants $M, \eta > 0$ exist such that, whenever $0 < |x| < \eta$, we have $|R(x)| \le M|x|^k$. The notation $O(d^{-k})$ concerns $d \to \infty$, with other parameters fixed. A subscript in O_d permits dependence on the fixed parameter d.

For $0 < |x| < 1$, integrate the geometric series:

$$\frac{\arctan x}{x} = 1 - \frac{x^2}{3} + \frac{x^4}{5} + O(x^6), \qquad \frac{\operatorname{artanh} x}{x} = 1 + \frac{x^2}{3} + \frac{x^4}{5} + O(x^6).$$

For example, the remainder in the second identity satisfies

$$\left| \sum_{k=3}^{\infty} \frac{x^{2k}}{2k+1} \right| \le \frac{|x|^6}{7} \sum_{j=0}^{\infty} |x|^{2j} = \frac{|x|^6}{7(1-x^2)}.$$

The absolute remainder of the first, alternating series is bounded by the same positive series. The series for p in Section 10 gives

$$p(x) = -x^5/3 + R_p(x), \qquad |R_p(x)| \le \frac{|x|^9}{5(1-|x|^4)}.$$

Put $b = \tanh 1$, $h = \tan 1$, $\delta_\beta = \beta - 1$ and $\delta_\alpha = \alpha - 1$. The first and second derivatives used below are

$$(\log\cosh y)' = \tanh y, \quad (\log\cosh y)'' = 1 - \tanh^2 y,$$

$$(-\log\cos y)' = \tan y, \quad (-\log\cos y)'' = 1 + \tan^2 y.$$

Differentiating once more,

$$(\log\cosh y)''' = -2\tanh y\ \operatorname{sech}^2 y, \qquad (-\log\cos y)''' = 2\tan y\ \sec^2 y.$$

The absolute value of the first expression is at most 2. The second needs to be bounded only between α and 1. Since $0 \le \alpha \le 1$, its absolute value is at most $2\tan 1\ \sec^2 1$. Write R_β, R_α for the two Taylor integral remainders. Then

$$|R_\beta| \le \frac{2}{6}|\delta_\beta|^3, \qquad |R_\alpha| \le \frac{2\tan 1\ \sec^2 1}{6}|\delta_\alpha|^3.$$

Here the factor $1/6$ comes from $\frac{1}{2}\int_0^{|\delta|}(|\delta| - t)^2 dt = |\delta|^3/6$. The expansions are

$$\log\cosh(1+\delta_\beta)-\log\cosh 1=b\delta_\beta+\frac{1-b^2}{2}\delta_\beta^2+O(\delta_\beta^3),$$

$$-\log\cos(1+\delta_\alpha)+\log\cos 1=h\delta_\alpha+\frac{1+h^2}{2}\delta_\alpha^2+O(\delta_\alpha^3).$$

Since

$$\delta_\beta=x^2/3+x^4/5+O(x^6),\quad \delta_\alpha=-x^2/3+x^4/5+O(x^6),$$

$$\delta_\beta^2=x^4/9+O(x^6),\quad \delta_\alpha^2=x^4/9+O(x^6),\quad \delta_\beta^3=O(x^6),\quad \delta_\alpha^3=O(x^6),$$

substitution gives

$$\log\cosh\beta-\log\cosh 1=\frac{b}{3}x^2+\left(\frac{b}{5}+\frac{1-b^2}{18}\right)x^4+O(x^6),$$

$$-\log\cos\alpha+\log\cos 1=-\frac{h}{3}x^2+\left(\frac{h}{5}+\frac{1+h^2}{18}\right)x^4+O(x^6).$$

Add $p(x)$ and put $L=\cosh 1/\cos 1$:

$$\log\frac{\mathfrak{r}(x)}{L}=a_2x^2+\ell_4x^4-\frac{x^5}{3}+O(x^6),$$

$$a_2=\frac{b-h}{3},\qquad \ell_4=\frac{b+h}{5}+\frac{2+h^2-b^2}{18}.$$

Denote the right-hand side by $z(x)$. Since $z=O(x^2)$,

$$e^z=1+z+\frac{z^2}{2}+O(z^3),\qquad \frac{z^2}{2}=\frac{a_2^2}{2}x^4+O(x^6),\quad z^3=O(x^6).$$

Therefore

$$\mathfrak{r}(x)=L\left[1+a_2x^2+\left(\ell_4+\frac{a_2^2}{2}\right)x^4-\frac{x^5}{3}+O(x^6)\right].$$

Define $a_4=\ell_4+a_2^2/2$ and substitute $x=1/d$:

$$\boxed{\mathfrak{r}_d=L\left[1+\frac{a_2}{d^2}+\frac{a_4}{d^4}-\frac{1}{3d^5}+O(d^{-6})\right].}$$

The coefficients of the first and third powers are zero; the fifth-power coefficient is $-L/3$. Numerical values for comparison are

$$La_2\approx -0.757603347512237,\qquad La_4\approx 2.035222101370122.$$

The variable in this expansion is the base order d. We fix d and take the block limit before letting d grow. In total order $N=dm$, this does not provide a uniform $O(N^{-2})$ extremal error.

Part II: The unique maximizer of fixed-base expectations

1. The precise extremal object

Definition 9 (The two fixed-base blow-up models). Fix an odd integer $d \geq 3$. For any order-d tournament T^*, replace each vertex by a random tournament of order m. All edges between blocks follow T^*. The total order is $N = dm$.

Write $\mathcal{T}(T^*, m)$ for independent uniform random internal blocks, and $\mathcal{R}(T^*, m)$ for independent uniform regular internal blocks; the latter requires odd m. The expected cycle and path counts are denoted by $\mathbb{E}C, \mathbb{E}H$ and normalized by

$$\nu_N = (N-1)!/2^N, \qquad \mu_N = N!/2^{N-1}.$$

Theorem 10 (The unique expected extremum). We prove that for each fixed d there is M_d such that whenever $m \geq M_d$, both models' cycle and path expectations are maximized exactly by bases ordinarily isomorphic to Car_d. In $\mathcal{R}$ only admissible m are considered.

This gives a proof manuscript for Conjecture 1 in the cited Wormald version and its path counterpart. The objects compared are expectations of random blow-up models, not individual samples, and no exact Hamiltonian extremum for all tournaments of order N is asserted.

2. The regular-base constants are already spectral ratios

For a regular base, Part I gives

$$\lim_{m\to\infty} \frac{\mathbb{E}C_{\mathcal{T}}}{\nu_N} = \lim_{m\to\infty} \frac{\mathbb{E}H_{\mathcal{T}}}{\mu_N} = \rho_d(S),$$

$$\lim_{\substack{m\to\infty \\ m \text{ odd}}} \frac{\mathbb{E}C_{\mathcal{R}}}{\nu_N} = \lim_{\substack{m\to\infty \\ m \text{ odd}}} \frac{\mathbb{E}H_{\mathcal{R}}}{\mu_N} = e^{1/d}\rho_d(S),$$

$$\rho_d(S) = \frac{\det(I + S/d)}{\det(I + iS/d)}.$$

Comparing regular bases is thus a finite matrix problem. Distinguish the algebraic definition from the probabilistic conclusion: the determinant ratio below can be defined for nonregular sign matrices, but this section invokes the equality between that ratio and a model expectation limit only for regular bases. The expectation for a nonregular base cannot use that limit formula; Sections 7-10 prove separately that its normalized expectation tends to zero.

3. Product comparison, integration and all equality conditions

Quoted theorem 11 (The Fang-Huang product comparison). We invoke Fang-Huang's Corollary 3.4. This is an external theorem, not a result reproved here. Its comparison matrix is

$$(K_d)_{ij} = \begin{cases} 1, & i+j \leq d, \\ 0, & i+j = d+1, \\ (-1)^{i+j-d-2}, & i+j \geq d+2. \end{cases}$$

Here p is any positive integer, $x \in \mathbb{R}^d$, and $S_1, \ldots, S_p$ are real skew-symmetric matrices of order d with off-diagonal entries in ± 1. The vector x^* lists the absolute values of the coordinates of x in decreasing order. For any product of these tournament sign matrices of the same order, the cited theorem gives

$$(S_1 \cdots S_p x)^* \prec_w K_d^p x^*.$$

The right-hand side is a nonnegative decreasing vector. Weak majorization $u \prec_w v$ means

$$\sum_{i=1}^{r} u_i \le \sum_{i=1}^{r} v_i \quad (1 \le r \le d).$$

The cited theorem also states that vector equality in the outermost comparison implies vector equality for every suffix product. We now prove the squared-norm comparison and the equality conversion needed here.

Lemma 12 (Truncated sums under weak majorization). Write $(a-t)_+ = \max(a-t, 0)$. For nonnegative decreasing vectors satisfying $u \prec_w v$ and $t \ge 0$,

$$\sum_{i=1}^{d} (u_i - t)_+ = \max_{0 \le r \le d} \left(\sum_{i=1}^{r} u_i - rt \right) \le \max_{0 \le r \le d} \left(\sum_{i=1}^{r} v_i - rt \right) = \sum_{i=1}^{d} (v_i - t)_+.$$

Proof. (Squared norms and equality)

For the first equality, take r equal to the number of coordinates strictly greater than t. Adding smaller coordinates can only decrease the sum, and removing larger coordinates can only decrease it. For any $a \ge 0$,

$$2 \int_0^\infty (a-t)_+ \, dt = 2 \int_0^a (a-t) \, dt = 2[at - t^2/2]_0^a = a^2.$$

Integrating coordinate by coordinate gives

$$\|u\|_2^2 = 2 \int_0^\infty \sum_i (u_i - t)_+ \, dt \le 2 \int_0^\infty \sum_i (v_i - t)_+ \, dt = \|v\|_2^2.$$

If the squared norms agree, set $g(t) = \sum_i (v_i - t)_+ - \sum_i (u_i - t)_+$. This function is continuous and nonnegative, with integral zero. If $g(t_0) > 0$, continuity supplies an interval of positive length on which $g(t) > g(t_0)/2$, making the integral positive. Thus $g(t) = 0$ for every $t \ge 0$. Right derivatives of these piecewise linear functions give

For a finite set, $\#U = |U|$ is its cardinality. The scalar indicator $\mathbf{1}_{\mathcal{E}}$ is 1 when condition $\mathcal{E}$ holds and 0 otherwise; it differs from the all-one vector $\mathbf{1}$ without a condition subscript.

$$\#\{i : u_i > t\} = \#\{i : v_i > t\} =: N(t).$$

For coordinate j,

$$u_j = \int_0^\infty \mathbf{1}_{\{\#\{i : u_i > t\} \ge j\}} \, dt = \int_0^\infty \mathbf{1}_{\{N(t) \ge j\}} \, dt = v_j.$$

Equality of squared norms therefore forces $u = v$. Taking $x = e_j$ and using $e_j^* = e_1$ yields

$$\|S^p e_j\|_2^2 \le \|K_d^p e_1\|_2^2.$$

Equality also forces $(S^p e_j)^* = K_d^p e_1$. □

4. Every moment of the comparison matrix matches the transitive matrix

Write $\operatorname{tr} M = \sum_j M_{jj}$.

Let T_d have upper-triangular entries 1, lower-triangular entries -1 and diagonal entries 0. Let R_0 be the reversal permutation matrix and $D_0 = \operatorname{diag}(1, -1, \ldots)$. Substituting the definition of K_d entry by entry gives

$$(D_0 K_d R_0)_{ij} = (-1)^{i-1}(K_d)_{i,d+1-j} = \begin{cases} (-1)^{i-1}, & i < j, \\ 0, & i = j, \\ (-1)^{j}, & i > j. \end{cases}$$

More explicitly, $R_0 e_j = e_{d+1-j}$ and $(D_0)_{jj} = (-1)^{j-1}$. Every entry of K_d depends only on $i + j$, so $(K_d)_{ij} = (K_d)_{ji}$ and $K_d^T = K_d$.

Write the sequence of increasing odd indices followed by decreasing even indices as

$$(\ell_1, \ldots, \ell_d) = (1, 3, \ldots, d, d-1, d-3, \ldots, 2), \qquad \Lambda e_j = e_{\ell_j}.$$

Each index occurs once. Hence $\Lambda^T \Lambda = \Lambda \Lambda^T = I$, and $\ell_1 = 1$.

Order both rows and columns by increasing odd indices followed by decreasing even indices. For two odd indices, the earlier one satisfies $i < j$, giving $(-1)^{i-1} = 1$. For two even indices, the earlier one satisfies $i > j$, giving $(-1)^j = 1$. Every odd index precedes every even index: if their original indices satisfy $i < j$, the entry is $(-1)^{i-1} = 1$; if $i > j$, it is $(-1)^j = 1$. All upper-triangular entries in this order are therefore 1. Let Λ perform this permutation. Then

$$T_d = \Lambda^T D_0 K_d R_0 \Lambda.$$

Set $U = D_0 \Lambda$. Since $R_0 R_0^T = I$, $\Lambda \Lambda^T = I$ and $K_d^T = K_d$, multiplication gives

$$\begin{aligned} T_d T_d^T &= \Lambda^T D_0 K_d R_0 \Lambda \Lambda^T R_0^T K_d D_0 \Lambda \\ &= \Lambda^T D_0 K_d^2 D_0 \Lambda = U^T K_d^2 U. \end{aligned}$$

The first index in this ordering is 1, so $U e_1 = e_1$. Also $T_d^T = -T_d$, hence

$$T_d^2 = -U^T K_d^2 U, \qquad T_d^{2p} = (-1)^p U^T K_d^{2p} U$$

because each adjacent pair $U U^T$ in the product equals I. Consequently,

$$(-1)^p (T_d^{2p})_{11} = e_1^T K_d^{2p} e_1 = (K_d^p e_1)^T (K_d^p e_1) = \|K_d^p e_1\|_2^2.$$

Appendix A.2 proves entry by entry that the shift Z with negative wraparound satisfies $Z^T T_d Z = T_d$. Inserting $Z Z^T = I$ gives $Z^T T_d^{2p} Z = T_d^{2p}$. Its diagonal entry at $j < d$ gives $(T_d^{2p})_{j+1,j+1} = (T_d^{2p})_{jj}$. At d, the two wraparound minus signs multiply to 1, giving $(T_d^{2p})_{11} = (T_d^{2p})_{dd}$. Thus

$$\begin{aligned} \|T_d^p\|_F^2 &= \operatorname{tr}((T_d^p)^T T_d^p) = (-1)^p \operatorname{tr} T_d^{2p} \\ &= d(-1)^p (T_d^{2p})_{11} = d\|K_d^p e_1\|_2^2. \end{aligned}$$

Sum the d column inequalities from the preceding section:

$$\|S^p\|_F^2 = \sum_{j=1}^{d} \|S^p e_j\|_2^2 \le d\|K_d^p e_1\|_2^2 = \|T_d^p\|_F^2 \qquad (p \ge 1).$$

This includes every odd p. We have not replaced a theorem solely about traces of degree $4k$ by one about degree $4k+2$.

5. The spectral-ratio bound and the full equality argument

Here S is any tournament sign matrix of order d, with no regularity assumption. In this section λ_j denotes a frequency of the unnormalized matrix S; to compare with the frequencies of $C = S/d$ in Part I, multiply those frequencies by d. Nonzero eigenvalues occur in pairs $\pm i\lambda_j$, with $\lambda_j > 0$. The Frobenius estimate in Part I uses only skew symmetry and off-diagonal entries in ± 1, not regularity, so each $x_j = \lambda_j^2/d^2 < 1/2$. Zero eigenvalues contribute 1 to the following determinants, while a nonzero two-dimensional block contributes $1 + x_j$ and $1 - x_j$, respectively. Therefore

$$\det(I + S/d) = \prod_j (1 + x_j), \qquad \det(I + iS/d) = \prod_j (1 - x_j) > 0.$$

For $0 \le x < 1$, integrate the geometric series:

$$\begin{aligned} \log \frac{1+x}{1-x} &= \int_0^x \left(\frac{1}{1+t} + \frac{1}{1-t} \right) dt \\ &= 2\int_0^x \frac{dt}{1-t^2} = 2\sum_{r=0}^{\infty} \int_0^x t^{2r}\, dt = 2\sum_{r=0}^{\infty} \frac{x^{2r+1}}{2r+1}. \end{aligned}$$

The interchange follows from uniform convergence of the geometric series on $[0, x]$. The skew-symmetric canonical form also gives $\|S^p\|_F^2 = 2\sum_j \lambda_j^{2p}$. Substituting term by term yields

$$\begin{aligned} \log \rho_d(S) &= 2\sum_j \sum_{r=0}^{\infty} \frac{\lambda_j^{4r+2}}{(2r+1)d^{4r+2}} \\ &= \sum_{r=0}^{\infty} \frac{\|S^{2r+1}\|_F^2}{(2r+1)d^{4r+2}} \le \sum_{r=0}^{\infty} \frac{\|T_d^{2r+1}\|_F^2}{(2r+1)d^{4r+2}} = \log \rho_d(T_d). \end{aligned}$$

Define $r_d = \rho_d(T_d)$. Every deficit is nonnegative. If their sum vanishes, the term $r = 1$ vanishes, giving $\|S^3\|_F^2 = \|T_d^3\|_F^2$. This deficit is a sum of d nonnegative column deficits, so each column attains equality. The squared-norm equality criterion in Section 3 of this part and Fang-Huang's suffix equality give, in sequence,

$$(S^3 e_j)^* = K_d^3 e_1, \qquad (S^2 e_j)^* = K_d^2 e_1.$$

Take squared norms in the second identity and sum over j:

$$\operatorname{tr} S^4 = \|S^2\|_F^2 = d\|K_d^2 e_1\|_2^2 = \|T_d^2\|_F^2 = \operatorname{tr} T_d^4.$$

Appendix A.3 independently proves that this equality forces $S = \Omega^T T_d \Omega$, where Ω is a signed permutation matrix. Check the converse as well: $\Omega^T\Omega = I$, so

$$I + tS = \Omega^T (I + tT_d)\Omega, \qquad \det(I + tS) = (\det \Omega)^2 \det(I + tT_d) = \det(I + tT_d).$$

Lemma 13 (The spectral ratio and equality). Take $t = 1/d$ and $t = i/d$ separately to obtain $\rho_d(S) = r_d$. Thus

$$\rho_d(S) \le r_d, \quad \text{with equality if and only if } S = \Omega^T T_d \Omega.$$

Proof. (Ordinary isomorphism in the regular case)

The signed permutation matrix Ω here has exactly one entry ± 1 in each row and column and satisfies $\Omega^T\Omega = I$. Write $\Omega = D_\varepsilon P$, where P is an ordinary permutation matrix and $D_\varepsilon = \operatorname{diag}(\varepsilon_1, \dots, \varepsilon_d)$; $\varepsilon_i \in \{-1, 1\}$ and $\varepsilon = (\varepsilon_1, \dots, \varepsilon_d)^T$. Then

$$S = P^T D_\varepsilon T_d D_\varepsilon P, \qquad PSP^T = D_\varepsilon T_d D_\varepsilon.$$

Only P is absorbed into ordinary relabeling; continue to denote the relabeled matrix by S. Diagonal sign switching itself is not an ordinary isomorphism. If S is regular, then after this relabeling $S\mathbf{1} = 0$ still holds. Using $D_\varepsilon^2 = I$,

$$D_\varepsilon T_d \varepsilon = 0, \qquad T_d \varepsilon = 0.$$

The difference between rows i and $i+1$ has entry 1 in each of columns $i, i+1$ and zero elsewhere. Therefore

$$0 = (T_d\varepsilon)_i - (T_d\varepsilon)_{i+1} = \varepsilon_i + \varepsilon_{i+1}.$$

Recursion gives $\varepsilon_i = (-1)^{i-1}\varepsilon_1$. The overall sign ε_1 occurs twice in $D_\varepsilon T_d D_\varepsilon$ and cancels. Appendix A.2 supplies an ordinary isomorphism, checked on every edge, between this alternating switching and Car_d. Among regular bases, equality therefore holds exactly for the ordinary carousel isomorphism class. □

6. An explicit strict gap from the sixth moment

$\binom{n}{k} = n!/(k!(n-k)!)$, with value 0 for $k > n$.

Lemma 14 (Strict sixth-moment deficit). If regular S is not ordinarily isomorphic to the carousel, the preceding equality argument gives

$$\Delta_3 := \|T_d^3\|_F^2 - \|S^3\|_F^2 > 0.$$

Proof. (An explicit gap)

Modulo 2, $S \equiv J + I$. Since $J^2 = dJ$ and $J^3 = d^2J$,

$$S^3 \equiv (J+I)^3 = I + 3J + 3dJ + d^2J \equiv I + (1 + d + d^2)J \equiv I + J \pmod 2.$$

The last step uses that $d(d+1)$ is even. Also $(S^3)^T = -S^3$, so its diagonal entries vanish and its off-diagonal entries are odd. If $z = 2k+1$,

$$z^2 = 4k(k+1) + 1 \equiv 1 \pmod 8.$$

Pair the skew-symmetric positions in the squared norm:

$$\|S^3\|_F^2 = 2\sum_{i<j}((S^3)_{ij})^2 \equiv 2\binom{d}{2} = d(d-1) \pmod{16}.$$

This also applies to T_d. Thus Δ_3 is a positive multiple of 16, and $\Delta_3 \geq 16$. Retaining just term $r = 1$ in the logarithmic series gives

$$\log \frac{r_d}{\rho_d(S)} \geq \frac{\Delta_3}{3d^6} \geq \frac{16}{3d^6}.$$

Exponentiate and rearrange:

$$\rho_d(S) \leq r_d e^{-16/(3d^6)}, \qquad r_d - \rho_d(S) \geq r_d(1 - e^{-16/(3d^6)}) > 0.$$

This is a gap between limiting constants for fixed bases, not an effective threshold for finite block order.

7. Nonregular scores and exponential loss

Definition 15 (Degree parameters for a nonregular base). Let d_i be the base outdegrees, put $b_i = d_i + 1/2$, and define

$$a_i = \frac{2b_i}{d}, \qquad \kappa = \left(\prod_{i=1}^{d} a_i\right)^{1/d}, \qquad s_i = 2d_i - (d-1).$$

Put $s = (s_1, \ldots, s_d)^T$, so that $\|s\|_2^2 = \sum_{i=1}^{d} s_i^2$. Each edge contributes exactly one outdegree. Therefore

$$\sum_i d_i = \binom{d}{2}, \qquad \sum_i a_i = \frac{2\binom{d}{2} + d}{d} = d, \qquad a_i = 1 + \frac{s_i}{d} \in (0, 2), \qquad \sum_i s_i = 0.$$

The function $f(a) = a - 1 - \log a$ satisfies $f'(a) = (a-1)/a$, so its unique minimum occurs at $a = 1$ with value $f(1) = 0$. Hence

$$\log \kappa = \frac{1}{d}\sum_i \log a_i \leq \frac{1}{d}\sum_i (a_i - 1) = 0.$$

Equality requires $a_i = 1$ for every index, equivalently $d_i = (d-1)/2$ for every index. For a nonregular base, $\kappa < 1$. To quantify the loss, use a second-order expansion with integral remainder:

$$\log a = (a-1) - (a-1)^2 \int_0^1 \frac{1-t}{[1 + t(a-1)]^2}\, dt.$$

Since $0 < 1 + t(a-1) < 2$,

$$\int_0^1 \frac{1-t}{[1+t(a-1)]^2}\, dt \geq \frac{1}{4}\int_0^1 (1-t)\, dt = \frac{1}{8}.$$

Substitute a_i and sum:

$$\log \kappa \leq \frac{1}{d}\sum_i \left(\frac{s_i}{d} - \frac{s_i^2}{8d^2}\right) = -\frac{\|s\|_2^2}{8d^3}.$$

For odd d each s_i is even. Nonregularity implies that some $s_i \neq 0$; the zero-sum condition then requires nonzero terms of opposite signs, each with absolute value at least 2. Thus $\|s\|_2^2 \geq 2^2 + 2^2 = 8$ and

$$\kappa^N \leq \exp[-N\|s\|_2^2/(8d^3)] \leq e^{-N/d^3}.$$

In model $\mathcal{T}$, let W be the entrywise expectation of the adjacency matrix. For any nonnegative matrix of order N, let $\mathrm{Sym}(N)$ be all vertex permutations and let $\mathcal{C}_N$ consist of permutations with a single cycle of length N. Define the weighted permanent and weighted Hamiltonian cycle count by:

$$\operatorname{per} W = \sum_{\sigma \in \mathrm{Sym}(N)} \prod_{v=1}^{N} W_{v,\sigma(v)}, \qquad C(W) = \sum_{\sigma \in \mathcal{C}_N} \prod_{v=1}^{N} W_{v,\sigma(v)}.$$

A single-cycle permutation already identifies cyclic starting points, so no additional division by the number of vertices is made. A fixed directed Hamiltonian cycle uses N distinct unordered vertex pairs, and its random internal edges are independent. Its occurrence probability is therefore the product of the corresponding entries of W. Sum over cycles identified up to cyclic starting point:

$$\mathbb{E}C_{\mathcal{T}} = C(W) \leq \operatorname{per} W.$$

The permanent contains all permutations. Each Hamiltonian cycle corresponds to one single-cycle permutation with nonnegative weight, proving the final inequality. We have not assumed independence for the edges in an arbitrary permutation: such a permutation can contain a two-cycle, whereas a tournament cannot contain both opposite edges.

8. Filling weighted rows, making them integral and applying Brégman

Write $\lceil a \rceil$ for the smallest integer at least a.

Definition 16 (Integral targets for row sums). We apply this section for $m \geq 2$. Model $\mathcal{R}$ also requires odd m, so there $m \geq 3$. This covers all sufficiently large block orders. Every row of type i in W has sum

$$r_i = md_i + \frac{m-1}{2} = mb_i - \frac{1}{2}, \qquad R_i = \lceil r_i \rceil.$$

For a row of length L, denoted w, assume $0 \leq w_j \leq 1$. Its current sum is $r = \sum_{j=1}^{L} w_j$, and its target integer sum R satisfies $r \leq R \leq L$. The amount to add is $s_0 = R - r$, and the available capacity is $\sum_j (1 - w_j) = L - r \geq s_0$. Starting at $j = 1$, choose successively

$$\delta_j = \min(1 - w_j, s_{j-1}), \qquad s_j = s_{j-1} - \delta_j, \qquad w'_j = w_j + \delta_j.$$

At each step the remaining mass satisfies

$$s_j = \max(0, s_{j-1} - (1 - w_j)) = \max\left(0, s_0 - \sum_{h=1}^{j} (1 - w_h)\right).$$

The first equality subtracts the smaller of two quantities; the second follows by substitution starting at $j = 1$.

The recursion gives $s_L = \max(0, s_0 - \sum_j (1 - w_j)) = 0$, so $0 \leq w'_j \leq 1$ and $\sum_j w'_j = R$. Every coefficient of the permanent is nonnegative, so these entrywise increases cannot decrease it.

Fix the other rows and write the permanent as $\sum_j w_j c_j$, where c_j is the permanent obtained by deleting the current row and column j; thus $c_j \geq 0$. If entries w_a, w_b both lie strictly between 0 and 1, label them so that $c_a \geq c_b$ and take

$$\eta = \min(1 - w_a, w_b) > 0, \qquad w'_a = w_a + \eta, \qquad w'_b = w_b - \eta.$$

The row sum is preserved, and

$$\sum_j w'_j c_j - \sum_j w_j c_j = \eta(c_a - c_b) \geq 0.$$

At least one coordinate becomes 0 or 1, reducing the number of nonintegral coordinates. After at most $L - 1$ steps there is at most one nonintegral coordinate. Since all other coordinates and the row sum R are integers, the remaining coordinate must also be integral. Perform this operation on every row to obtain a zero-one matrix $W^\sharp$. Each row of type i has R_i ones, and $\operatorname{per} W \leq \operatorname{per} W^\sharp$.

Here $W^\sharp$ is an auxiliary matrix for the upper bound. Its required properties are zero-one entries and the specified row sums; diagonal entries may be 1. The theorem below applies to such general zero-one matrices.

Now invoke Brégman's inequality, as stated in Lemma 2.1 of Alon's paper: for a zero-one matrix with row sums $q_1, \ldots, q_L > 0$,

$$\operatorname{per} W^\sharp \leq \prod_{\ell=1}^{L} (q_\ell!)^{1/q_\ell}.$$

Since $r_i \geq (m-1)/2 > 0$, the integer upper bound R_i is at least 1. This verifies the positive-row-sum hypothesis of the cited Brégman bound. There are exactly m rows of each type here, giving

$$\operatorname{per} W \leq \prod_{i=1}^{d} (R_i!)^{m/R_i}.$$

This uses the original zero-one theorem; the concrete row operations above prove its applicability to the weighted matrix.

9. Normalization with explicit constants

If the permanent being estimated is zero, the nonnegative cycle count it bounds is also zero, and the desired bound already holds; we do not take the logarithm of zero. All logarithmic permanent calculations below and in Section 10 are made only when the corresponding permanent is positive. The resulting positive upper bounds also cover the zero case. Assume $m \geq 2$ and $N = dm$. Write $R_i = mb_i + \xi_i$. Then $|\xi_i| \leq 1/2$ and

$$R_i \geq m/2 - 1/2 \geq m/4, \qquad R_i \leq N.$$

The rounding error follows from the target definition:

$$0 \leq R_i - r_i < 1, \qquad \xi_i = R_i - mb_i = (R_i - r_i) - \frac{1}{2} \in [-1/2, 1/2).$$

Use the remainder form of Stirling's bound for integers $R \geq 1$:

$$\log(R!) = R \log R - R + \frac{1}{2} \log(2\pi R) + \frac{\theta_R}{12R}, \qquad 0 < \theta_R < 1.$$

Therefore

$$m\sum_{i=1}^{d}\frac{\log(R_i!)}{R_i}\le m\sum_i\log R_i-N+m\sum_i\frac{\log(2\pi R_i)}{2R_i}+m\sum_i\frac{1}{12R_i^2}\le m\sum_i\log R_i-N+2d\log(2\pi N)+\frac{4d}{3m}.$$

For the last two terms we used $m/(2R_i)\le 2$ and $m/(12R_i^2)\le 4/(3m)$, respectively. The lower Stirling bound also gives

$$\log(N!/2^N)\ge N\log(N/2)-N.$$

Subtracting yields

$$\log\frac{\operatorname{per}W}{N!/2^N}\le m\sum_i\log\frac{2R_i}{N}+2d\log(2\pi N)+\frac{4d}{3m}.$$

Set $y_i=\xi_i/(mb_i)$. Because $b_i\ge 1/2$, we have $|y_i|\le 1/m\le 1/2$ and

$$\frac{2R_i}{N}=a_i(1+y_i),\qquad |\log(1+y_i)|=\left|\int_0^{y_i}\frac{dt}{1+t}\right|\le\frac{|y_i|}{1-|y_i|}\le\frac{2}{m}.$$

Thus

$$m\sum_i\log\frac{2R_i}{N}\le m\sum_i\log a_i+2d=N\log\kappa+2d.$$

Use $4d/(3m)\le 2d/3$ and $N!/2^N=N\nu_N$ to obtain the explicit bound

$$\frac{\mathbb{E}C_{\mathcal{T}}}{\nu_N}\le e^{8d/3}(2\pi)^{2d}N^{2d+1}\kappa^N.$$

With regular internal blocks, the actual old row sum in each sample is the positive integer $mb_i-1/2$. For a sample G with adjacency matrix A_G, each cycle is a nonnegative single-cycle term in the permanent, so $C(G)\le\operatorname{per}A_G$. Apply Brégman directly to A_G, use the preceding estimates, and average over the samples to obtain

$$\frac{\mathbb{E}C_{\mathcal{R}}}{\nu_N}\le e^{8d/3}(2\pi)^{2d}N^{2d+1}\kappa^N.$$

The regular model uses only admissible odd m. For a nonregular base, $\kappa<1$. Take the logarithm of the right-hand side and divide by N:

$$\frac{\log(e^{8d/3}(2\pi)^{2d}N^{2d+1}\kappa^N)}{N}=\log\kappa+\frac{8d/3+2d\log(2\pi)+(2d+1)\log N}{N}\longrightarrow\log\kappa<0,$$

For completeness, for $N\ge 1$,

$$0\le\frac{\log N}{N}=\frac{1}{N}\int_1^N\frac{dt}{t}\le\frac{1}{N}\int_1^N\frac{dt}{\sqrt{t}}=\frac{2(\sqrt{N}-1)}{N}\longrightarrow 0.$$

Denote the upper bound by B_N and set $c = -\frac{1}{2}\log\kappa > 0$. Since $\log B_N/N \to \log\kappa = -2c$, beyond a threshold we have $\log B_N/N < -c$, and therefore $0 \le B_N < e^{-cN} \to 0$. The polynomial degree in the path bound differs, but division of its logarithm by N again makes the constant terms and $\log(N+1)/N$ vanish by the displayed integral estimate.

Both normalized cycle expectations therefore tend to zero.

10. The added-vertex path bound with every normalization term retained

Lemma 17 (Random augmentation and path counts). Fix the old tournament G. Add a vertex v whose edges to old vertices are independently and fairly oriented. Each path $x_1 \to \cdots \to x_N$ corresponds to the unique candidate cycle $v \to x_1 \to \cdots \to x_N \to v$. The two required new edges have joint probability $(1/2)^2 = 1/4$. Deleting v reverses the correspondence. Therefore

$$\mathbb{E}[C_{\text{aug}} \mid G] = H(G)/4, \qquad 4\nu_{N+1} = \frac{4N!}{2^{N+1}} = \frac{N!}{2^{N-1}} = \mu_N,$$

$$\frac{\mathbb{E}H}{\mu_N} = \frac{\mathbb{E}C_{\text{aug}}}{\nu_{N+1}}.$$

Here G_{aug} is the random tournament after adding v, and $C_{\text{aug}} := C(G_{\text{aug}})$. Conditional expectation first fixes G and averages over the added edges. For independent internal blocks let W_{aug} be the entrywise expected augmented adjacency matrix. For regular internal blocks fix an actual augmented sample and denote its adjacency matrix by A_{aug}. Apply the row filling and integrality operations of Section 8 to either input. Denote the resulting zero-one auxiliary matrix of order $(N+1)$ by $W^{\sharp}_{\text{aug}}$; it need not be a tournament adjacency matrix.

$$\mathbb{E}C_{\text{aug},\mathcal{T}} = C(W_{\text{aug}}) \le \operatorname{per} W_{\text{aug}} \le \operatorname{per} W^{\sharp}_{\text{aug}},$$

$$C(G_{\text{aug}}) \le \operatorname{per} A_{\text{aug}} \le \operatorname{per} W^{\sharp}_{\text{aug}} \qquad (\text{each sample of } \mathcal{R}).$$

The first equality is used only for independent internal blocks. The second line holds separately for each sample and does not require independent edges inside a regular block.

For the augmented expectation matrix of $\mathcal{T}$, the old row sum is mb_i. For each actual augmented sample of $\mathcal{R}$, the old row sum is at most $mb_i + 1/2$. Use the common choice

$$R'_i = \lceil mb_i + 1/2 \rceil = mb_i + \xi'_i, \qquad \frac{1}{2} \le \xi'_i < \frac{3}{2}.$$

Fill the new row to $R_0 = N$. Since $b_i \le d - 1/2$ and $m \ge 2$, we have $mb_i + 1/2 \le N - 1/2$ and hence $R'_i \le N$. Also $R'_i \ge m/2$. These filling targets do not exceed the augmented matrix's column count $N+1$, so the row operations of Section 8 apply. Brégman gives

$$\log \operatorname{per} W^{\sharp}_{\text{aug}} \le m \sum_i \frac{\log(R'_i!)}{R'_i} + \frac{\log(N!)}{N}.$$

Use $R'_i \ge m/4$ and $R'_i \le N+1$ to estimate the remainders for the old rows, keeping the new row separate:

$$\begin{aligned}\log \operatorname{per} W^{\sharp}_{\mathrm{aug}} &\le m\sum_i \log R'_i - N + \log N - 1 \\ &\quad + 2d\log(2\pi(N+1)) + \frac{4d}{3m} + \frac{\log(2\pi N)}{2N} + \frac{1}{12N^2} \\ &\le m\sum_i \log R'_i - N + \log N - 1 \\ &\quad + (2d+1)\log(2\pi(N+1)) + \frac{4d}{3m} + \frac{1}{12}.\end{aligned}$$

Subtract $\log((N+1)!/2^{N+1}) \ge (N+1)\log((N+1)/2) - (N+1)$. Regroup every leading term as follows:

$$\begin{aligned}&m\sum_i \log R'_i + \log N - (N+1)\log((N+1)/2) \\ &= m\sum_i \log(2R'_i/N) + N\log(N/(N+1)) + \log(2N/(N+1)).\end{aligned}$$

The second term is at most zero, and the third is at most $\log 2$. The inequality $\log(1+y) \le y$, valid for $y \ge 0$, follows from the integral $\int_0^y (1+t)^{-1}dt \le y$. Thus

$$\begin{aligned}m\sum_i \log(2R'_i/N) &= N\log\kappa + m\sum_i \log(1+\xi'_i/(mb_i)) \\ &\le N\log\kappa + \sum_i \frac{\xi'_i}{b_i} \le N\log\kappa + 3d.\end{aligned}$$

Using $4d/(3m) \le 2d/3$ now gives

$$\begin{aligned}\log \frac{\operatorname{per} W^{\sharp}_{\mathrm{aug}}}{(N+1)!/2^{N+1}} &\le N\log\kappa + \frac{11d}{3} + \frac{1}{12} + \log 2 \\ &\quad + (2d+1)\log(2\pi(N+1)).\end{aligned}$$

Exponentiate and substitute $(N+1)!/2^{N+1} = (N+1)\nu_{N+1}$. The two models satisfy, respectively,

$$\frac{\mathbb{E}H_{\mathcal{T}}}{\mu_N} \le 2e^{11d/3+1/12}(2\pi)^{2d+1}(N+1)^{2d+2}\kappa^N,$$

$$\frac{\mathbb{E}H_{\mathcal{R}}}{\mu_N} \le 2e^{11d/3+1/12}(2\pi)^{2d+1}(N+1)^{2d+2}\kappa^N.$$

For the second bound, constrain the augmented cycle count in each sample and then take expectations. No independence of edges inside regular blocks is assumed. For a fixed nonregular base, the logarithm of either bound divided by N tends to $\log\kappa < 0$, so both bounds tend to zero.

11. From limiting constants to the exact expected extremum for large blocks

Proof. (A common threshold and uniqueness)

For fixed d, each vertex pair has two possible directions, so there are $2^{\binom{d}{2}}$ labeled bases and at most that many ordinary isomorphism classes. Among regular bases, the carousel has limiting constant r_d, or $e^{1/d}r_d$, and all other regular bases have the positive gap from Section 6. The normalized expectations for nonregular bases tend to zero.

For each other base choose sufficiently large m so that its actual normalized expectation is strictly smaller than the carousel's. Take the maximum of the finitely many thresholds to obtain M_d from Section 1 and the unique ordinary isomorphism class. This uses fixed d and does not assert uniformity as d grows.

It remains to explain why every ordinarily isomorphic copy of the carousel attains the same maximum. Given an ordinary base isomorphism π, write a blow-up vertex as (i, a). Then

$$(i, a) \longmapsto (\pi(i), a)$$

is a bijection between samples of the two blow-up models. The base isomorphism preserves directions between blocks; internal blocks are merely reordered, preserving their independent identical sampling probabilities. Every path and cycle corresponds uniquely under this bijection, so their expectations agree. Combined with the strict inequality for nonisomorphic bases, this identifies the entire ordinary carousel isomorphism class as the equality class. □

12. Conclusions and scope

Part I gives a finite algebraic proof for Conjecture 6 and the limit and monotonicity in Conjecture 2 of the cited Wormald version. With explicit use of his counting theorems and Fang-Huang's comparison, this manuscript obtains the fixed-base expectation assertion in Conjecture 1 and its path counterpart.

The conclusions concern model expectations with fixed base order. They do not establish Wormald's Conjecture 3 for all tournaments, an effective threshold M_d, or a simultaneous limit with growing base and block orders.

Appendices: Auxiliary calculations

A.1 The determinant polynomial of the transitive sign matrix

Let T_n have upper-triangular entries 1, lower-triangular entries -1 and diagonal entries 0, and write $F_n(t) = \det(I + tT_n)$. We prove the closed form of Part I, Section 9 directly by row operations, without treating even principal determinants as an unexplained input.

For $n = 1$, $F_1(t) = 1 = ((1 + t) + (1 - t))/2$, proving the formula in that case. Now assume $n \geq 2$. For the first $n - 1$ rows simultaneously perform $R_i \leftarrow R_i - R_{i+1}$. The operation is implemented by an upper-triangular matrix with diagonal 1, so the determinant is preserved. In the resulting matrix the only potentially nonzero entries in the first $n - 1$ rows are

$$M_{ii} = 1 + t, \qquad M_{i,i+1} = t - 1$$

The last row is $(-t, \ldots, -t, 1)$. Temporarily assume $t \neq -1, 0$ and put

$$r = \frac{1 - t}{1 + t}.$$

Denote the top-left block of order $(n - 1)$ by B. Its determinant is $(1 + t)^{n-1}$. The top-right column b has only its last entry nonzero, equal to $t - 1$. Solve $Bx = b$ from the last row upwards:

$$x_{n-1} = \frac{t - 1}{1 + t} = -r,$$

$$(1 + t)x_i + (t - 1)x_{i+1} = 0 \quad \Longrightarrow \quad x_i = rx_{i+1} = -r^{n-i}.$$

The Schur complement gives

$$F_n(t) = (1+t)^{n-1}\left(1 - t\sum_{k=1}^{n-1} r^k\right).$$

First multiply the finite geometric sum by $1-r$; adjacent terms cancel:

$$(1-r)\sum_{k=1}^{n-1} r^k = (r + r^2 + \cdots + r^{n-1}) - (r^2 + \cdots + r^n) = r - r^n.$$

Thus $\sum_{k=1}^{n-1} r^k = r(1-r^{n-1})/(1-r)$. Using $1-r = 2t/(1+t)$, the expression in parentheses becomes

$$\begin{aligned} 1 - t\sum_{k=1}^{n-1} r^k &= 1 - \frac{tr(1-r^{n-1})}{1-r} \\ &= 1 - \frac{1-t}{2}(1-r^{n-1}) \\ &= \frac{1+t}{2} + \frac{1-t}{2} r^{n-1}. \end{aligned}$$

Multiply the bracket by $(1+t)^{n-1}$ and substitute $r = (1-t)/(1+t)$:

$$\begin{aligned} F_n(t) &= \frac{(1+t)^n}{2} + \frac{(1+t)^{n-1}(1-t)}{2}\left(\frac{1-t}{1+t}\right)^{n-1} \\ &= \frac{(1+t)^n}{2} + \frac{(1-t)^n}{2}. \end{aligned}$$

Both sides are polynomials and agree for every t except two values. They therefore also agree at $t = -1, 0$. Substitute $t = 1/d$ and $t = i/d$ to obtain the determinant ratio for the carousel model.

To justify the assertion about each principal determinant in Part I, Section 9, take nonempty U and put $k = |U|$. In increasing original vertex order, $T_d[U]$ is exactly T_k. In $F_k(t) = \det(I + tT_k)$, the coefficient of t^k is $\det T_k$; write $[t^k]$ for extraction of this coefficient. Thus

$$\det T_d[U] = \det T_k = [t^k]F_k(t) = \frac{1+(-1)^k}{2} = \begin{cases} 1, & k \text{ even}, \\ 0, & k \text{ odd}. \end{cases}$$

The empty principal determinant is 1 by convention. Thus the proved closed form determines every individual principal determinant.

A.2 Alternating switching and ordinary carousel isomorphism

Let $d = 2q+1$. Switch the transitive matrix by alternating signs $\varepsilon_i = (-1)^{i-1}$ and $D_\varepsilon = \operatorname{diag}(\varepsilon_i)$. For $i < j$ and $\Delta = j - i$,

$$(D_\varepsilon T_d D_\varepsilon)_{ij} = (-1)^\Delta.$$

Define the carousel labeling map

$$\phi(i) = (q+1)i \pmod d.$$

Since $2(q+1) = d+1$, the numbers $q+1$ and d are coprime, making ϕ a permutation.

If $\Delta = 2r$, then $1 \le r \le q$ and

$$\phi(j) - \phi(i) \equiv (q+1)2r \equiv r \pmod{d}.$$

This difference belongs to $\{1, \dots, q\}$, precisely the condition for $\phi(i) \to \phi(j)$ in the carousel. The switched matrix entry is also 1.

If $\Delta = 2r + 1$, then $0 \le r \le q - 1$ and

$$\phi(j) - \phi(i) \equiv q + 1 + r \pmod{d}.$$

This difference belongs to $\{q+1, \dots, 2q\}$, so the direction is reversed; the switched matrix entry is -1. Skew symmetry handles $i > j$, and all diagonal entries are zero. Thus the map preserves each directed edge and is an ordinary isomorphism.

Part II also uses equality of the diagonal entries of powers of the transitive matrix. Define $Ze_j = e_{j+1}$ for $j < d$ and $Ze_d = -e_1$. When $i, j < d$,

$$(Z^T T_d Z)_{ij} = (T_d)_{i+1,j+1} = (T_d)_{ij}.$$

When $i < d, j = d$,

$$(Z^T T_d Z)_{i,d} = -(T_d)_{i+1,1} = 1 = (T_d)_{i,d}.$$

When $i = d, j < d$,

$$(Z^T T_d Z)_{d,j} = -(T_d)_{1,j+1} = -1 = (T_d)_{d,j}.$$

The remaining diagonal entry is zero in both matrices. Therefore $Z^T T_d Z = T_d$, and for every integer $p \ge 0$,

$$Z^T T_d^{2p} Z = T_d^{2p}.$$

Taking diagonal entries makes adjacent diagonal entries equal, with the two wraparound minus signs canceling.

A.3 Complete expansions for the fourth-order determinant and fourth-moment equality

Let a, b, c, g, e, f be six local real scalars, with g distinct from the matrix order. Set

$$M = \begin{pmatrix} 0 & a & b & c \\ -a & 0 & g & e \\ -b & -g & 0 & f \\ -c & -e & -f & 0 \end{pmatrix}.$$

Write M_{1j} for the order-three determinant obtained by deleting row 1 and column j, without the cofactor sign. Expanding along the first row gives the following separate calculations:

$$
\begin{aligned}
M_{12} &= \det \begin{pmatrix} -a & g & e \\ -b & 0 & f \\ -c & -f & 0 \end{pmatrix} \\
&= (-a)(0 - f(-f)) - g((-b)0 - f(-c)) + e((-b)(-f) - 0(-c)) \\
&= -af^2 - cgf + bef, \\
M_{13} &= \det \begin{pmatrix} -a & 0 & e \\ -b & -g & f \\ -c & -e & 0 \end{pmatrix} \\
&= (-a)((-g)0 - f(-e)) - 0 + e((-b)(-e) - (-g)(-c)) \\
&= -aef + be^2 - cge, \\
M_{14} &= \det \begin{pmatrix} -a & 0 & g \\ -b & -g & 0 \\ -c & -e & -f \end{pmatrix} \\
&= (-a)((-g)(-f) - 0(-e)) - 0 + g((-b)(-e) - (-g)(-c)) \\
&= -agf + bge - cg^2.
\end{aligned}
$$

Each order-three determinant is itself expanded along its first row. Its three contributions come from the three first-row entries, with a zero entry contributing zero. Multiply by the coefficients in the order-four expansion:

$$
\begin{aligned}
\det M &= -aM_{12} + bM_{13} - cM_{14} \\
&= (a^2f^2 + acgf - abef) + (-abef + b^2e^2 - bcge) + (acgf - bcge + c^2g^2) \\
&= a^2f^2 + b^2e^2 + c^2g^2 - 2abef + 2acgf - 2bcge \\
&= (af - be + cg)^2.
\end{aligned}
$$

When all six variables are ± 1, the expression $af - be + cg$ is a sum of three numbers in ± 1, taking values ± 1 or ± 3. Thus the determinant is 1 or 9. Let $D_4(S)$ count four-vertex principal determinants equal to 9. Expand by column multilinearity, choosing in each column either the identity column or the corresponding column of tS:

$$
\det(I + tS) = \sum_{U \subseteq [d]} t^{|U|} \det S[U].
$$

An odd-order skew-symmetric matrix B satisfies $\det B = \det B^T = \det(-B) = -\det B$, so its determinant vanishes. Every second-order principal determinant is 1. Among fourth-order determinants, D_4 are 9 and all others are 1. Therefore

$$
\det(I + tS) = 1 + e_2 t^2 + e_4 t^4 + O(t^6),
$$

$$
e_2 = \binom{d}{2}, \qquad e_4 = (\binom{d}{4} - D_4) + 9D_4 = \binom{d}{4} + 8D_4.
$$

Use the following expansion only near $t = 0$. Choose $|t|\,\|S\|_{\mathrm{op}} < 1/2$. Every eigenvalue μ then satisfies $|t\mu| < 1/2$. Define $\log(1 + t\mu)$ by the logarithmic series with value 0 at 1. Interchanging the power series with a finite eigenvalue sum is valid, and

$$\det(I+tS)=\prod_{\mu}(1+t\mu),\qquad \operatorname{tr}S^k=\sum_{\mu}\mu^k.$$

Eigenvalues are counted with multiplicity. The real skew-symmetric canonical form gives diagonalizability over the complex numbers; zero eigenvalues are included.

The logarithmic series for each eigenvalue gives

$$\log\det(I+tS)=\sum_{k\ge1}\frac{(-1)^{k+1}t^k}{k}\operatorname{tr}S^k=-\frac{t^2}{2}\operatorname{tr}S^2-\frac{t^4}{4}\operatorname{tr}S^4+O(t^6).$$

Odd traces vanish because $\operatorname{tr}S^{2j+1}=\operatorname{tr}(S^{2j+1})^T=-\operatorname{tr}S^{2j+1}$. On the other hand, substitute $z=e_2t^2+e_4t^4+O(t^6)$ into $\log(1+z)=z-z^2/2+O(z^3)$. Since $z^2=e_2^2t^4+O(t^6)$ and $z^3=O(t^6)$, this gives

$$\log\det(I+tS)=e_2t^2+(e_4-e_2^2/2)t^4+O(t^6).$$

Equate the fourth-order coefficients and isolate the trace step by step:

$$-\operatorname{tr}S^4/4=e_4-e_2^2/2,\qquad \operatorname{tr}S^4=2e_2^2-4e_4=2\left(\binom{d}{2}\right)^2-4\binom{d}{4}-32D_4(S).$$

In the transitive matrix, any four vertices in increasing order have $a=b=c=g=e=f=1$, giving determinant $(1-1+1)^2=1$. Thus $D_4(T_d)=0$. Subtracting yields

$$\operatorname{tr}T_d^4-\operatorname{tr}S^4=32D_4(S).$$

If the fourth moments agree, $D_4(S)=0$. Choose $\varepsilon_1=1,\varepsilon_j=s_{1j}$ for $j>1$, and define $D_\varepsilon=\operatorname{diag}(\varepsilon_1,\dots,\varepsilon_d)$ and $S'=D_\varepsilon SD_\varepsilon$. Then $s'_{1j}=\varepsilon_1s_{1j}\varepsilon_j=s_{1j}^2=1$. Each principal determinant is multiplied by $(\det D_\varepsilon[U])^2=1$, so D_4 is unchanged. Choose any $i<j<k$ all different from 1:

$$\det S'[\{1,i,j,k\}]=(s'_{jk}-s'_{ik}+s'_{ij})^2.$$

If these three vertices form a directed triangle, $s'_{ij}=s'_{jk}=-s'_{ik}$, making the expression in parentheses $3s'_{ij}$ and its square 9, a contradiction. Thus the remaining vertices contain no directed triangle. If $i\to j$ and $j\to k$ but $i\not\to k$, then $k\to i$, producing a triangle. The direction relation is therefore a strict total order, and we can label vertices in this order. Vertex 1 dominates all remaining vertices, making the entire switched tournament transitive. Conversely, signed permutation similarity preserves traces, so it gives fourth-moment equality. For $d=3$, two vertices remain after switching; order them according to their unique edge.

A.4 Reproducible steps for row integrality

Definition 18 (A termination count for rounding). Section 8 of Part II lists the coordinate filling amounts δ_j, the remaining amounts s_j, the two-coordinate movement η, and the permanent change. Here are two integers that establish termination.

$$q(w)=\#\{j:0<w_j<1\},\qquad q(W)=\sum_{\ell=1}^{L}q(W_{\ell,\cdot}).$$

Each two-coordinate movement preserves the integer row sum, makes at least one nonintegral coordinate integral, and leaves other integral coordinates unchanged. Thus $q(w)$ decreases by at least 1. Initially $q(w) \le L$, so it cannot decrease indefinitely. If it stopped at $q(w) = 1$, the unique nonintegral coordinate would satisfy

$$w_a = R - \sum_{j \neq a} w_j \in \mathbb{Z},$$

contradicting its nonintegrality. Thus termination occurs at $q(w) = 0$. Processing a row does not change other rows. Every row becomes integral after at most $L-1$ two-coordinate movements, and the total number is at most $L(L-1)$. At every step the permanent changes by $\eta(c_a - c_b) \ge 0$, as computed in Section 8.

A.5 Checking each regrouped term in the added-vertex bound

Section 10 of Part II already gives the path bound with explicit constants. Here we split its logarithmic regrouping into separate terms so that no additional factor N is hidden. Since $md = N$,

$$m \sum_i \log R_i' = m \sum_i \log(2R_i'/N) + N \log(N/2).$$

We obtain

$$\begin{aligned} &N \log(N/2) + \log N - (N+1)\log((N+1)/2) \\ &= N \log N - N \log 2 + \log N - (N+1)\log(N+1) + (N+1)\log 2 \\ &= N[\log N - \log(N+1)] + [\log N + \log 2 - \log(N+1)] \\ &= N \log(N/(N+1)) + \log(2N/(N+1)). \end{aligned}$$

The denominator in the ensuing exponential bound is initially $(N+1)!/2^{N+1}$. On changing to the cycle-expectation denominator,

$$\frac{(N+1)!/2^{N+1}}{\nu_{N+1}} = \frac{(N+1)!}{N!} = N+1.$$

The polynomial degree changes from $2d+1$ to $2d+2$. The added-vertex identity also gives

$$\frac{\mathbb{E}H}{\mu_N} = \frac{4\mathbb{E}C_{\text{aug}}}{4\nu_{N+1}} = \frac{\mathbb{E}C_{\text{aug}}}{\nu_{N+1}}.$$

One factor comes from the change in order and the other from the added-vertex probability; both have been retained explicitly.

A.6 A common threshold for finitely many bases

Definition 19 (Normalized functions on isomorphism classes). Here T denotes an ordinary isomorphism class of base tournaments; choose any representative to form the random blow-up. Part II, Section 11 proved that the expectation is independent of that choice. For a model $\mathcal{M} \in \{\mathcal{T}, \mathcal{R}\}$ and a fixed choice of count, define

$$f_T(m) = \begin{cases} \mathbb{E}C_{\mathcal{M}}(T,m)/\nu_{dm}, & \text{cycles}, \\ \mathbb{E}H_{\mathcal{M}}(T,m)/\mu_{dm}, & \text{paths}, \end{cases} \qquad f_*(m) = f_{\text{Car}_d}(m).$$

Write $\ell_T = \lim_{m\to\infty} f_T(m)$ and $\ell_* = \lim_{m\to\infty} f_*(m)$. In model $\mathcal{R}$, both limits are restricted to odd m.

Fix one model and one count. Let the carousel limit be $\ell_* > 0$ and let every other ordinary isomorphism class have limit $\ell_T < \ell_*$. Since $d \geq 3$, the transitive base is nonregular whereas the carousel is regular, so the set of noncarousel isomorphism classes is nonempty. Finiteness and isomorphism invariance here are supplied by Part II, Section 11. Each noncarousel limit is strictly smaller, so the finite set of positive differences has a positive minimum:

$$\delta = \min_{T \not\cong \mathrm{Car}_d} (\ell_* - \ell_T) > 0.$$

Nonregular classes have limit zero; regular noncarousel classes are handled by the strict spectral gap. For each class choose M_T so that whenever $m \geq M_T$ its normalized expectation $f_T(m)$ satisfies

$$|f_T(m) - \ell_T| < \delta/3.$$

Impose the same error requirement on the carousel. Let M be the maximum of these finitely many thresholds. Then

$$f_*(m) - f_T(m) > (\ell_* - \delta/3) - (\ell_T + \delta/3) \geq \delta/3 > 0.$$

Finally take the maximum of the four thresholds for two models and two counts. The regular internal-block model uses only odd m. This proves existence without inventing a numerical threshold from an ineffective asymptotic theorem.

A.7 A calculus proof of the trigonometric inequality used in monotonicity

Part I uses $\tan y \geq y + y^3/3$ for $0 \leq y < \pi/2$. Set $f(y) = \tan y - y$. Then

$$f'(y) = \tan^2 y \geq 0, \qquad f(0) = 0.$$

Thus $\tan y \geq y$. Next set $g(y) = \tan y - y - y^3/3$. We have

$$g'(y) = \tan^2 y - y^2 \geq 0, \qquad g(0) = 0.$$

The desired inequality follows. Moreover,

$$\alpha(x) = \int_0^1 \frac{dt}{1 + x^2 t^2} \geq \int_0^1 (1 - x^2 t^2)\, dt = 1 - x^2/3,$$

where $(1 + z)^{-1} - (1 - z) = z^2/(1 + z) \geq 0$. This supplies the two elementary estimates used in the monotonicity calculation.

References

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3. Noga Alon. *The maximum number of Hamiltonian paths in tournaments*. Combinatorica 10 (1990), 319-324. Author's full text. We use the Brégman permanent bound stated in Section 2.